Equilibrium or Retrieval?

Do LLM Agents Derive or Recall Game-Theoretic Equilibria,
and What That Means for Mechanism Design

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Abstract

Large language models (LLMs) play textbook games well. A model can reach the textbook answer in two ways, however: by *deriving* it from the rules, or by *recognising* a famous game and recalling its answer. The two are behaviourally identical on the textbook question, and they come apart only when the question changes. We separate them with *isomorphic twins*. Every question is asked in its standard wording and in a disguised wording with identical numbers, rules and information structure but no game vocabulary. A third, pre-registered study adds an *extreme* disguise: no story, coded answers and a neutral answer box. Every answer is paired with a competence probe in the same wording, so a change in play cannot be blamed on misreading. Four models from four providers answer 18,768 questions across a Vickrey auction, a Cournot duopoly, a finitely repeated prisoner's dilemma and a public goods game, under three pre-registrations.

The models differ sharply. Nemotron 3 Ultra and Kimi K3 play identically across the standard twins. In the extreme disguise, Nemotron's recognition of the Cournot game falls from 96% to 10% while its equilibrium play stays at 99%, which is direct evidence of derivation. Gemini 3.5 Flash Lite shows the opposite signature. It plays the Cournot equilibrium in 96% of standard questions but in only 4% of disguised ones, despite passing every competence probe, and its prisoner's-dilemma play flips from 91% cooperation to 70% defection once the game's name is gone. Recognition does not only retrieve equilibria: for the prisoner's dilemma it retrieves a cultural script of cooperation. [TBD: gpt-oss-120b; Stage 2B traps; Stage 2C for the remaining models.]

We then ask what noisy, quantal-response agents imply for mechanism design. We prove three results:

- a payment multiplier in a second-price auction is irrelevant for logit-QRE bidders but ruinous for bidders who recall "bid your value", which makes it a test of retrieval;
- the revenue-optimal reserve price is U-shaped in bidder precision but gains at most about 1% over Myerson's;
- the cooperation subsidy that achieves any target cooperation rate has a closed form, whose stage-game halfway point is free of parameters across the entire regular-QRE family.

A pre-registered test pits the derived subsidy against a classically guessed one: [TBD: Stage 5 result].

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1 Introduction

LLM agents are beginning to act in settings that game theory was built to analyse: bidding in auctions, setting prices, negotiating, and drawing on shared resources. The rules of such settings are designed, and designing them, which is the business of mechanism design, requires a model of how participants behave. Classical mechanism design assumes rational players who best-respond. Behavioural mechanism design replaces them with an empirically fitted model, most prominently the logit quantal response equilibrium (QRE) [6, 11], in which players choose better actions more often than worse ones, with a precision λ.

Before a fitted model of LLM play can be trusted to design anything, a prior question must be settled: *is the play we measure reasoning about the rules in front of the model, or retrieval of answers to famous problems?* The concern is concrete. The Vickrey auction, the Cournot duopoly, the prisoner's dilemma and the public goods game are among the most written-about games in any corpus a model is trained on. "Bid your true value", "produce $(a - c)/3b$" and "defect in the last round" appear thousands of times. LLM accuracy is known to depend on how familiar a task looks, not only on what it requires [10]. A mechanism tuned to recalled play may fail precisely when the designer changes the rules, which is the only reason to design a mechanism at all.

Approach. We hold each game fixed and vary only its *wording*:

(i) **Isomorphic twins.** The disguised wording preserves every number, rule and information structure, and removes the game's vocabulary. An extreme disguise (Stage 2C) removes the

story too, codes the answer (for example, "your offer is $3x + 5$") and replaces the answer box. Section 3 shows that a purely deriving agent's play is invariant to such relabelling, so any wording effect rejects pure derivation.

(ii) **Competence probes.** A short calculation in the same wording follows every answer. Only answers whose probe is passed are analysed, so a wording effect cannot be a reading failure.

(iii) **Measured recognition.** A validated code records whether the model's reasoning names the game (agreement with blind hand coding $\kappa = 1.00$). Recognition and play are then read jointly, by rules fixed before the data.

(iv) **Pre-registration.** Three registrations on OSF fix the predictions, sample sizes, exclusions and tests before each study [13]. The analysis code was locked by cryptographic fingerprint.

Findings so far.

- **Nemotron and Kimi** show no disguise effect in any game. The extreme disguise hides Cournot from Nemotron (recognition 96%→10%) without changing its play (98%→99%).
- **Gemini** shows a large retrieval signature: Cournot equilibrium play falls from 96% to 4% under disguise, with every competence probe passed.
- **In the prisoner's dilemma, recognition pulls *away* from equilibrium.** Named, Gemini cooperates in 91% of first rounds; disguised, it defects in 70%. Removing all game cues raises Nemotron's defection from 59% to 93%.

Contributions.

1. A design that identifies derived versus recalled strategic play: isomorphic twins, same-wording competence probes, measured recognition, and an extreme disguise, all pre-registered.
2. Evidence that the answer differs by model and by game, including a retrieval signature in Gemini's Cournot play and recognition-driven cooperation in the prisoner's dilemma.
3. Theory for mechanism design with QRE agents: an invariance theorem that turns a payment multiplier into a test of retrieval; the U-shaped revenue-optimal reserve; and a closed-form cooperation subsidy with a parameter-free prediction. These are tested on the same models in a pre-registered study.

2 Related work

LLMs as economic and strategic agents. Horton et al. [8] proposed treating LLMs as simulated economic agents and reproduced classic behavioural-economics findings with them. Akata et al. [2] placed LLMs in finitely repeated 2×2 games and found strong play in self-interested games, such as the prisoner's dilemma, and weak coordination. Fan et al. [5] tested basic requirements of rationality. Benchmarks such as GTBench [4] measure strategic reasoning across many game types. These studies vary the game; we vary only the wording of a fixed game, which isolates recognition from ability.

Behavioural game theory for LLMs. Jia et al. [9] evaluate 22 LLMs with behavioural models rather than Nash benchmarks. Pechon-Elkins and Chun [14] audit QRE-based measures of LLM strategic reasoning and conclude that fitted parameters are payoff- and model-dependent summaries rather than structural measures of rationality. We adopt that caution explicitly:

- we never compare λ across games, whose payoff scales differ;
- our main comparisons hold payoffs fixed by construction;
- we verify by parameter recovery that our fitting recovers known λ;
- we judge fit by held-out calibration, decided before the data.

Familiarity, memorisation and reasoning. McCoy et al. [10] show that LLM performance tracks how probable a task and its output are under the training distribution. Our twins apply this idea to strategic play, and the competence probes separate unfamiliarity of the *wording* from failure to *understand* it.

Mechanism design and bounded rationality. Our auction results build on the second-price auction [16] and on Myerson's optimal auction and implementability results [12], in the treatment of Roughgarden [15]. QRE was introduced by McKelvey and Palfrey [11] and is developed in Goeree et al. [6]. Our repeated-game discussion uses Benoît and Krishna [3].

3 A framework for derivation and retrieval

Let a *question* be a pair (G, w) of a game G (players, actions, payoffs, information) and a wording w that presents it. Two wordings w, w' are *isomorphic* for G if there is a bijection ϕ between their answer spaces such that, for every answer a, the payoff consequences of a under w equal those of $\phi(a)$ under w'. For example, "offer $= 3x + 5$" is isomorphic to "bid b" under $b = \phi(x) = 3x + 5$.

Definition 1. A model *derives* on G if the distribution of its answers depends on (G, w) only through G and the answer map: $\Pr(a \mid G, w') = \Pr(\phi^{-1}(a) \mid G, w)$ for every pair of isomorphic wordings. It *retrieves* if its answer distribution depends on features of w that are payoff-irrelevant, such as a game's name.

Proposition 1 (invariance)**.** *For a deriving model, every statistic of play that is defined through payoffs (for example the equilibrium-play rate, or λ fitted on payoffs) is identical across isomorphic wordings. Hence a difference in such a statistic across isomorphic twins, among answers whose competence probe is passed, rejects pure derivation.*

Proof. The statistic is a functional of the answer distribution mapped into payoff space. Under derivation, the mapped distributions coincide by definition. □

Remark. The converse fails. Invariance is also produced by a model that recalls an answer and recognises the game in *both* wordings. So equal play across twins is evidence of derivation only when the disguise actually hides the game. This is why we measure recognition and add an extreme disguise. Our pre-registered interpretation key reads "play unchanged, and recognition drops" as *consistent with* derivation, since recognition is coded from what a model writes, and silent recognition cannot be excluded.

A second route to identification changes the *rules* instead of the wording. If a trap variant has a different equilibrium from the famous game (for example, the winner pays twice the second bid), a retrieving model gives the famous answer where it is wrong. Theorem 1 shows that, for QRE players, this particular change is payoff-irrelevant up to scale, which sharpens the test.

One Cournot question, identical numbers, three wordings; each answer is followed by a maths check in the same wording

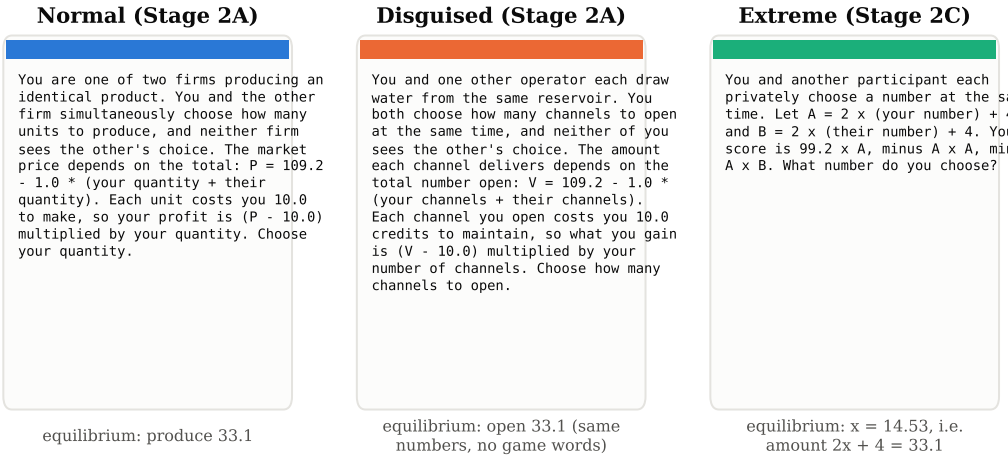


Figure 1: One Cournot question in its three wordings. All three versions have identical numbers and the same equilibrium (in the extreme wording, the coded answer decodes to the same amount). Each answer is followed by a maths check posed in the same wording.

Table 1: The four games. Parameters are drawn per question from seeded samplers.

Game	Parameters	Equilibrium	Disguised as
Vickrey auction	3 bidders, private value $v \sim U[10, 100]$	bid v	research stations requesting slots
Cournot duopoly	$P = a - Q$, $a \in [80, 120]$, $c = 10$	$q = (a - c)/3$	operators opening water channels
Repeated PD	$(T, R, P, S) = (5, 3, 1, 0)$, 10 rounds, round 1	defect	technicians setting panels X/Y
Public goods	n, endowment e, $1 < M < n$	contribute 0	operators routing signal

4 Games, wordings and competence probes

Games. Table 1 lists the four games. Each has a unique equilibrium action:

- a weakly dominant strategy in the auction;
- the Cournot–Nash quantity;
- defection by backward induction, since the horizon is common knowledge;
- zero contribution, the dominant strategy, in public goods.

Numbers are drawn per question from seeded samplers, so every question can be reproduced exactly.

Wordings. Each wording has three paraphrases (reworded, or reordered), which never add or drop a fact, so that a disguise effect can be told apart from a wording effect.

- The **disguised** wording moves the game into an unfamiliar story and keeps the numbers, the rules and who knows what.
- The PD's disguised labels `X`/`Y` are counterbalanced, so the cooperative label is randomised.

- The **extreme** wording (Stage 2C) has:
  - no story;
  - coded answers (Vickrey offer $= 3x + 5$ with a private prize; Cournot amount $= 2x + 4$ with the profit written in expanded form);
  - public goods asking how much to *keep*;
  - PD payoffs transformed to $2u - 4$;
  - a neutral answer box.

Each extreme question shares its seed and paraphrase index with a standard question, so every question exists in three wordings with identical numbers (Figure 1).

Competence probes. The probe is a one-step calculation in the question's own wording, for example "you request 50 and the largest other request is 60; what is your net result?". Verdicts:

- AT_EQUILIBRIUM: the probe is passed and the answer is within ± 0.5 of the equilibrium;
- STRATEGIC_GAP: the probe is passed and the equilibrium is missed;
- COMPETENCE_GAP: the probe is failed;
- parse and format failures.

Only the first two count. Server refusals and empty replies are re-asked, never scored.

Models and studies. gpt-oss-120b (via Groq), Nemotron 3 Ultra (NVIDIA), Kimi K3 (Moonshot, served by NVIDIA) and Gemini 3.5 Flash Lite (Google) were each queried with default sampling, September–November 2026. Each model answers:

Study	Registration	Questions per model
Stage 2A (named vs disguised)	osf.io/2uqpj	4 games × 2 wordings × 138 = 1,104
Stage 2B (traps, sweep, endgame)	osf.io/6hz8d	3,036
Stage 2C (extreme disguise)	osf.io/cqnsy	4 games × 138 = 552
Stage 5 (subsidy test)	pre-registration 4	300–400

The cell size of 138 gives 80% power for a 15-point change with Holm correction across four games.

5 Pre-registration and analysis

Registration and locks. Each pre-registration was fingerprinted (SHA-256) in git and registered on OSF before its data existed. Each fixes:

- the predictions;
- the exact sample sizes, as a stopping rule;
- exclusions;
- the tests and multiple-comparison corrections.

The analysis code is fingerprinted too, and it refuses to run on real data unless its fingerprint matches. Ten deviations, all recorded before the affected analysis, are listed in Appendix A. None changes a registered test.

Tests.

- **Stage 2A:** equilibrium play, named vs disguised, by two-proportion z-tests with Holm correction [7]. Public goods, the predicted null, uses an equivalence test (TOST, ±13 points).
- **Stage 2C:** each extreme question is paired with its twins; exact McNemar tests, Agresti–Min intervals for paired differences [1], and pooling by summing discordant pairs (the Mantel–Haenszel test for matched pairs).

Quantal response fits. In a logit QRE, a player chooses action a with probability $\propto \exp(\lambda\, u(a))$:

- **PD and Cournot:** u is computed at a symmetric fixed point;
- **public goods:** $u(g) = (M/n - 1)g$;
- **Vickrey:** rivals are assumed to bid their equilibrium bids, with values uniform on $[10, 100]$.

We compute the posterior over a log-uniform grid $\lambda \in [10^{-3}, 10^{3}]$. A cell whose evidence keeps rising towards the top of the grid gets only a lower bound, and is flagged as a ceiling. The headline is $\Delta\lambda = \lambda_{\text{named}} - \lambda_{\text{disguised}}$. In a parameter recovery on simulated QRE data, the nominal 95% intervals covered the true λ in 93.75% of 320 fits.

Reasoning codes. A locked codebook codes the stated reasoning. NAMES_GAME requires the reasoning to use a name for the game that the question does not use. Agreement with blind hand coding was $\kappa = 1.00$ on Stage 2A wording and 40/40 on a blind sample in the extreme wording. Error codes for missed answers were validated in two hand-coding rounds:

- COOPERATION_NORM: $\kappa = 0.75$;
- fraction slip: $\kappa = 0.72$;
- one code ("knew and overrode") failed validation ($\kappa = 0.49$) and is not used.

The fraction slip. With a number-only box, a model sometimes derives a fraction (683/60) and writes only part of it (683). A mechanical correction was fixed in pre-registration 3, before any Stage 2C data. It reads the answer as n/d when two conditions hold:

1. the box holds n, or n and d written together, for a fraction n/d in the model's own reasoning;
2. the reasoning ends on that fraction.

It is applied identically to both stages in the Stage 2C comparison, with uncorrected results reported alongside. Pre-registration 1's own analysis is reported uncorrected, as registered.

6 Results

6.1 Standard twins (Stage 2A)

Table 2 and Figure 2 give the registered test. Three patterns stand out.

Two models are invariant. Nemotron and Kimi play the same in both wordings in every game:

- both are at or near 100% in the auction and public goods;
- both are at 90–100% in Cournot;
- public goods is confirmed equivalent within ±13 points (TOST $p < 10^{-30}$ and $p = 2 \times 10^{-6}$).

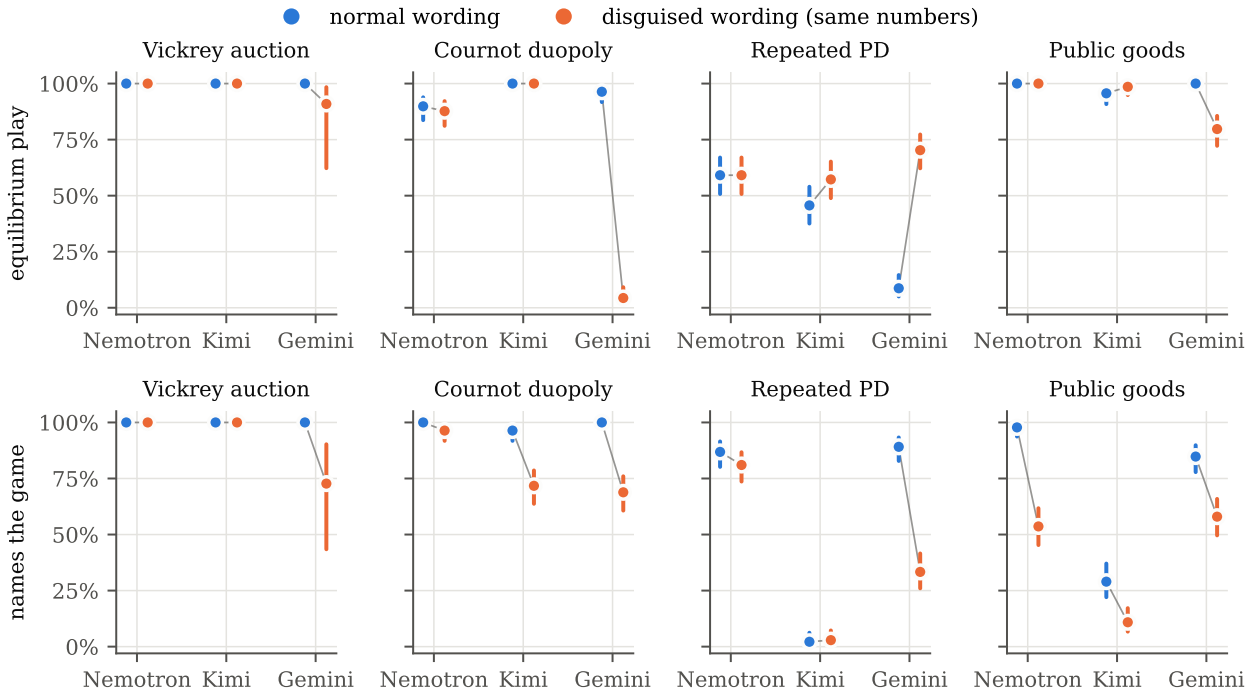


Figure 2: Stage 2A, normal vs disguised wording, with identical numbers. Top: equilibrium play among probe-passed answers. Bottom: the share of answers whose reasoning names the game. Bars are 95% Wilson intervals. Gemini's disguised Vickrey cell has $n = 11$: the model failed the disguised competence probe on 127 of 138 questions. [TBD: gpt-oss-120b, collection ends in late October.]

By Proposition 1 this is consistent with derivation. It is not proof, however: both models name the disguised game often (Figure 2, bottom). Nemotron names it in 96% of disguised Cournot answers and in 100% of disguised auctions. For these models the standard disguise does not hide the game, which motivates Stage 2C.

Gemini shows a retrieval signature in Cournot. Gemini plays the Cournot equilibrium in 96% of named questions and in 4% of disguised ones, a 92-point drop, yet passes the disguised competence probe in all 138 questions. It can compute profits in the disguised wording; it does not find the equilibrium there. Its disguised misses are mostly coded as a claimed but incorrect Nash calculation (44) or the collusive (joint-monopoly) quantity (31), so without the name it reasons towards the group optimum or mis-derives. The precision fits agree: λ falls from 0.649 $[0.50, 0.82]$ to 0.0047 $[0.0036, 0.0060]$, $\Delta\lambda = +0.645$, $P(\Delta\lambda > 0) = 1.00$. This is the pattern pre-registration 1 predicted for every model (a Cournot drop). It appears in one model of three.

The auction drop is a reading failure, not a strategic one. Gemini's disguised Vickrey cell keeps only 11 answers: it failed the disguised probe in 127 of 138 questions. The design classifies this correctly, as a competence gap, so it does not count as evidence about strategy. Without the probes, it would have looked like a 92-point strategic collapse.

In the prisoner's dilemma, recognition pulls away from equilibrium.

Table 2: Stage 2A main test (pre-registration 1): equilibrium play among probe-passed answers, named vs disguised. Holm correction across four games within each model.

Model	Game	Named	Disguised	Change	Holm p
Nemotron	Vickrey	100% (138/138)	100% (138/138)	+0	1
	Cournot	90% (124/138)	88% (121/138)	−2	1
	Repeated PD	59% (81/137)	59% (81/137)	+0	1
	Public goods	100% (136/136)	100% (138/138)	+0	1
Kimi	Vickrey	100% (138/138)	100% (138/138)	+0	1
	Cournot	100% (138/138)	100% (138/138)	+0	1
	Repeated PD	46% (63/138)	57% (79/138)	+12	0.22
	Public goods	96% (132/138)	99% (136/138)	+3	0.45
Gemini	Vickrey	100% (138/138)	91% (10/11)	−9	4×10^{-4}
	Cournot	96% (132/137)	**4% (6/138)**	**−92**	6×10^{-52}
	Repeated PD	9% (12/138)	**70% (97/138)**	**+62**	4×10^{-25}
	Public goods	100% (138/138)	80% (110/138)	−20	5×10^{-8}
gpt-oss-120b	all		[TBD: in collection]		

Table 3: Stage 2C (pre-registration 3), Nemotron: equilibrium play, normal (2A) vs extreme (2C), on paired questions with identical numbers. Exact McNemar test, Holm across games.

Game	Pairs	Normal	Extreme	Normal − extreme [95% CI]	Holm p
Vickrey	138	100%	99%	+1 [−1, +4]	1
Cournot	138	98%	99%	−1 [−5, +2]	1
Repeated PD	137	59%	93%	−34 [−43, −25]	3×10^{-10}
Public goods	136	100%	99%	+1 [−1, +3]	1

Recognition, disguised (2A) → extreme (2C): Vickrey 100%→100%, Cournot 96%→**10%**, PD 81%→65%, public goods 54%→99%.

- **Gemini:** with the game named, it cooperates in round 1 of a known 10-round game in 91% of questions; disguised, it defects in 70%. Every one of its 126 named-wording cooperative answers gives a cooperation-norm reason (trust, mutual benefit, establishing cooperation).
- **Kimi:** it moves the same way, by +12 points (Holm $p = 0.22$).

Recognition retrieves a *cultural* script, the folk wisdom of cooperation in repeated dilemmas, rather than the backward-induction equilibrium. Pre-registration 1 predicted this direction (equilibrium play rises under disguise), and named Gemini as the model with the largest swing. Both predictions held.

Public goods. Gemini contributes more when the game is disguised (equilibrium play 100%→80%; equivalence not shown). Most of its disguised misses argue for the group optimum. Kimi shows the small effect that pre-registration 1 hedged for: a slightly lower named-wording precision, $\Delta\lambda = -1.51$, 95% interval $[-2.36, -0.73]$.

6.2 The extreme disguise (Stage 2C)

The extreme disguise was built because the standard one did not hide the games from strong models. For Nemotron (Table 3, Figure 3):

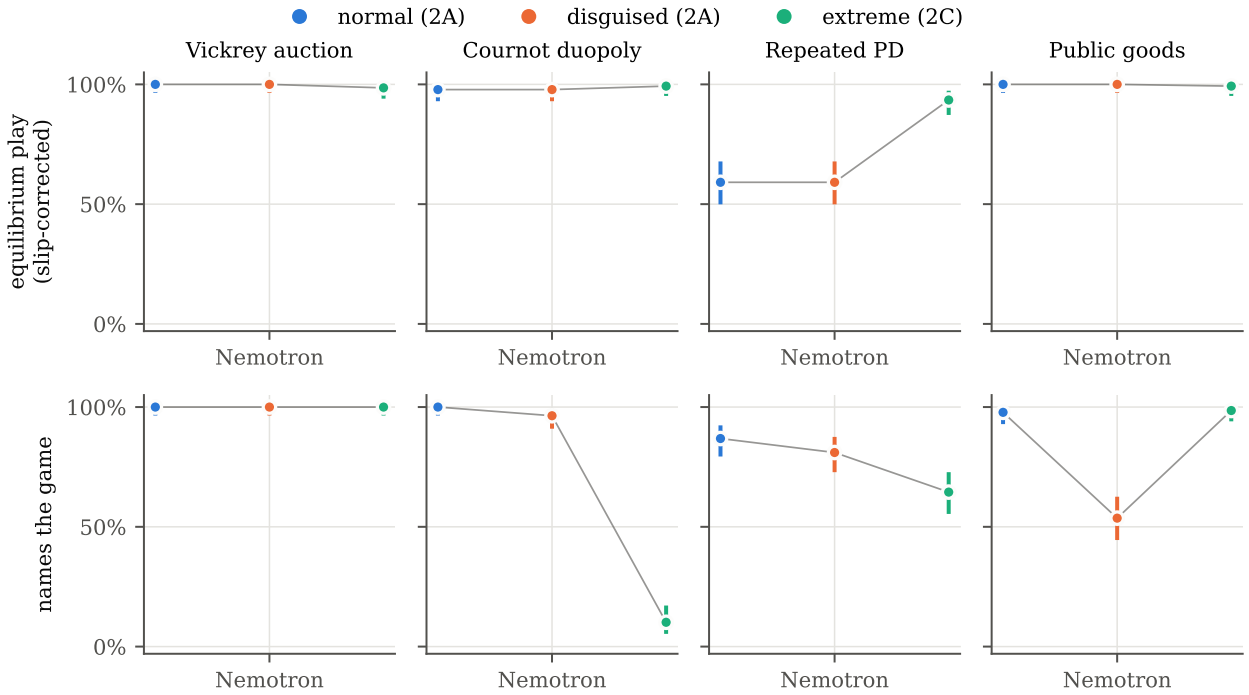


Figure 3: The same questions in three wordings, for each model whose Stage 2C is complete. Top: equilibrium play (fraction-slip corrected, both stages alike). Bottom: recognition. For Nemotron, the extreme disguise removes recognition of the Cournot game (96%→10%) and leaves play unchanged. [TBD: the remaining models.]

- **Cournot:** recognition falls from 96% to 10% ($p = 3 \times 10^{-36}$), while equilibrium play stays at 99%. Under the pre-registered interpretation key this is *consistent with derivation*: the model solves the game it no longer names. Within the extreme wording, play is the same whether or not the game is named (100% vs 99%).
- **Prisoner's dilemma:** defection rises from 59% to 93% (McNemar $p = 6 \times 10^{-11}$), and λ from 0.154 to 2.59. As with Gemini's standard disguise, stripping the familiar frame removes the cooperation pull.
- **Vickrey and public goods:** the disguise fails. Recognition stays at 99–100%, because the rules themselves identify the games. Play is unchanged, which the key classifies as inconclusive for derivation.

The fraction slip was not a detail. In the extreme wording, the coded answers are usually awkward fractions, so the slip becomes common: 50% of Nemotron's extreme Vickrey answers and 55% of its Cournot answers were written as a numerator or a concatenation. Without the correction fixed in advance, the comparison would report spurious drops of 51 and 46 points (both $p < 10^{-18}$). With it, the drops vanish. This illustrates why the scoring rule was registered before the data.

Pre-registered predictions. Pre-registration 3 predicted, among other things:

- a Cournot drop of at least 35 points for Nemotron: *not met* (−1);
- recognition of 20–35% for Cournot: *not met* (10%, which is lower);

- recognition of 45–65% for the PD: met (65%);
- flat public goods: met.

The author's predicted mechanism, that Nemotron recalled Cournot, is contradicted by the data.

6.3 Stage 2B: traps, number sweeps and the endgame

[TBD: Block B (traps): correct share and the recall signature (the textbook answer where it is wrong), named vs disguised, CMH across models. Block C (sweep): the wording×size interaction. Block E (endgame): defection at rounds 5, 9 and 10. Nemotron's Blocks B and C are complete; the registered analysis runs once all four models are.]

7 Mechanism design for noisy agents

We now treat each player as a logit-QRE player with precision λ and ask how rules should change. All results are verified numerically: the code accompanies the paper, and every claim has a test.

7.1 Second-price auctions with a reserve and a payment multiplier

Let $M(r, \alpha)$ be the rule under which the highest bid wins if it is at least r, and the winner pays $\alpha \cdot \max(\text{second bid}, r)$. The Vickrey auction is $M(0, 1)$, and under $M(0, \alpha)$ the dominant strategy is $b = v/\alpha$. In a symmetric logit QRE, each bidder's bid density is $\propto \exp(\lambda U(b \mid v))$, where U is the expected payoff against the other bidders' QRE bids.

Theorem 1 (α is irrelevant for QRE bidders)**.** *If $(B_1, \ldots, B_n)$ is a QRE of $M(r, 1)$ on bid space $[0, \bar{B}]$, then $(B_1/\alpha, \ldots, B_n/\alpha)$ is a QRE of $M(r/\alpha, \alpha)$ on $[0, \bar{B}/\alpha]$, with the same winner, the same revenue and the same payoffs.*

Proof. Against a highest rival bid H, $U_\alpha(b \mid v) = \mathbb{E}[(v - \alpha \max(H, r/\alpha))\mathbf{1}\{H < b,\ b \geq r/\alpha\}]$. Suppose rivals play $H = H_1/\alpha$, and substitute $b = \beta/\alpha$. Then $U_\alpha(\beta/\alpha \mid v) = \mathbb{E}[(v - \max(H_1, r))\mathbf{1}\{H_1 < \beta,\ \beta \geq r\}] = U_1(\beta \mid v)$. So the M_α bid has density $\propto \exp(\lambda U_1(\alpha b \mid v))$. By a change of variables, αB has density $\propto \exp(\lambda U_1(x \mid v))$, which is the M_1 QRE density, and the normalising constants agree because both integrate to one. The scaled profile is therefore self-consistent, which makes it a QRE. Dividing every bid by α preserves their order, and revenue is $\alpha \cdot B_{(2)}/\alpha = B_{(2)}$. □

The invariance needs the bid space to scale with α. Our solver lays the bid grid out in value units and confirms the identity to 10^{-9}.

A test of retrieval. For a deriving QRE bidder, $\alpha \cdot b$ has the same distribution at every α. A bidder that recalls "bid your value" bids v regardless. Under $\alpha = 2$, three such bidders leave the winner with $\mathbb{E}[v_{(1)}] - 2\,\mathbb{E}[v_{(2)}] = 77.5 - 110 = -32.5$. Any dependence of $\alpha \cdot b$ on α therefore cannot be QRE noise; it is the fingerprint of recall. Stage 2B's "winner pays twice" trap implements exactly this test: [TBD: trap result].

Theorem 2 (a reserve never raises efficiency)**.** *For any λ, expected efficiency (the winner's value, or 0 without a sale) cannot be increased by $r > 0$.*

Proof. For fixed bids, r never changes who has the highest bid; it can only cancel the sale, replacing a value of at least 10 by 0. Under QRE the bids also respond to r. That strategic effect is verified numerically: efficiency falls in r at every λ tested. □

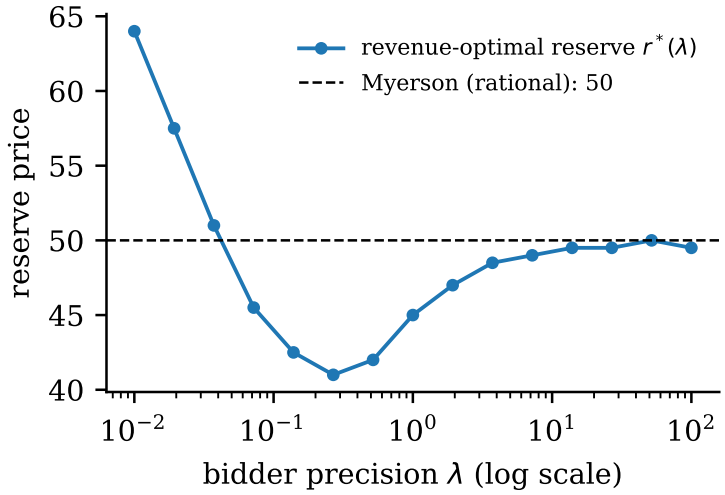


Figure 4: The revenue-optimal reserve for three logit-QRE bidders (values uniform on $[10, 100]$). It is U-shaped in precision and returns to Myerson's 50 as $\lambda \to \infty$.

The originally planned problem, maximising efficiency over (r, α), therefore has a trivial answer. For *revenue*, with values uniform on $[10, 100]$, rational bidders have $\varphi(v) = 2v - 100$, so Myerson's reserve is $r^* = 50$. For QRE bidders we solve the full symmetric fixed point, in which each bidder responds to the others' noisy bids, and optimise r (Figure 4).

The optimal reserve $r^*(\lambda)$ is U-shaped:

- **Moderate noise** ($\lambda \approx 0.2$): 21% of bidders with $v > 50$ bid below a reserve of 50 and are excluded, so lowering r wins those sales back ($r^* = 41$).
- **Extreme noise** ($\lambda \approx 0.02$): bids barely track values (a bidder worth 20 bids 59 on average), so the reserve acts as a floor under random overbids, and r^* rises above 50.

The revenue gain over Myerson's reserve never exceeds about 1%. Moreover, all three models analysed so far bid at the ceiling of precision in the standard auction. For these agents the auction leaves little to design, so we turn to the game where they are genuinely noisy.

7.2 A cooperation subsidy for the prisoner's dilemma

Under $S(\tau)$, every player who cooperates is paid an extra τ per round. Against an opponent who cooperates with probability p, defecting pays more by $\Delta U(p) = p(T-R)+(1-p)(P-S)$. Under $S(\tau)$ the advantage becomes $\Delta U(p) - \tau$, and the symmetric logit QRE satisfies $p = 1/(1 + e^{\lambda(\Delta U(p)-\tau)})$.

Theorem 3 (closed-form subsidy)**.** *The subsidy at which the symmetric logit-QRE cooperation rate equals p^* is*

$$\tau(p^*, \lambda) = \Delta U(p^*) + \frac{\operatorname{logit}(p^*)}{\lambda}.$$

Since $d\tau/dp = (T - R) - (P - S) + 1/(\lambda p(1 - p)) > 0$, each τ gives exactly one QRE.

Proof. Rearranging, $(1 - p)/p = e^{\lambda(\Delta U(p)-\tau)}$, so $-\operatorname{logit}(p) = \lambda(\Delta U(p) - \tau)$. Solve for τ. □

Corollary 1 (a parameter-free halfway point)**.** *In the stage game, at $\tau_{1/2} = \Delta U(\frac{1}{2}) = \frac{1}{2}[(T - R) + (P - S)]$ (that is, 1.5 for our payoffs), every regular QRE population cooperates with probability exactly $\frac{1}{2}$, whatever its precision and whatever its symmetric error distribution.*

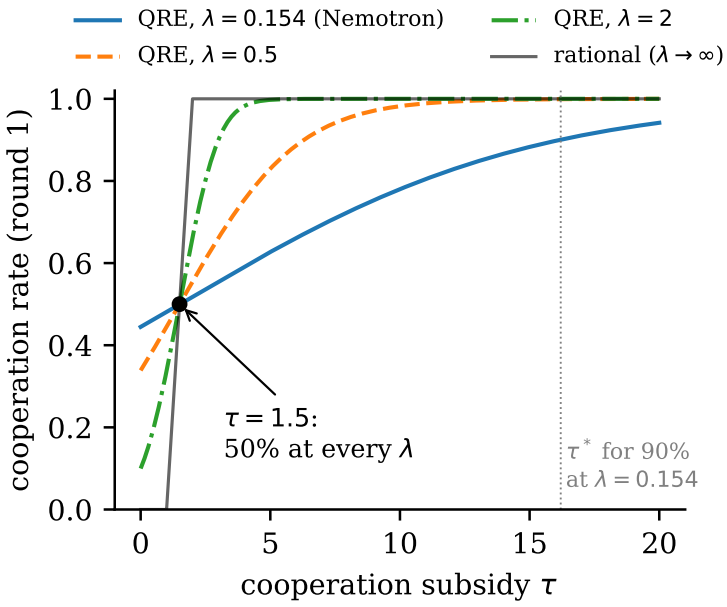


Figure 5: Cooperation under a subsidy τ for logit-QRE players. Every precision passes through 50% at $\tau = 1.5$ (Corollary 1), as does the rational mixed equilibrium. The dotted line is Theorem 3's subsidy for 90% cooperation at Nemotron's fitted $\lambda = 0.154$.

Proof. At $p = \frac{1}{2}$ and $\tau = \tau_{1/2}$ the player is indifferent, so a regular quantal responder randomises evenly. Uniqueness holds because $\Delta U(p) - \tau$ rises in p, while the response falls in it. □

Rational players and the repeated game.

- **Rational stage game:** players defect for $\tau < P - S = 1$ and cooperate for $\tau > T - R = 2$. In between, the subsidised stage game is chicken, whose symmetric equilibrium mixes with $p = \tau - 1$, so it too passes through $\frac{1}{2}$ at $\tau = 1.5$.
- **Repeated game:** chicken has two pure equilibria that pay each player differently, so in the 10-round game, round-1 cooperation can be subgame-perfect: play (C, C), and punish a deviator in the last round with the equilibrium worst for them [3]. At $\tau = 1.5$, deviating gains 0.5, while the punishment costs at least 1.5. So cooperation above 50% at $\tau = 1.5$ is consistent with rational repeated-game play as well as with a cooperation norm, and the reasoning codes separate the two.

Noisy players are cheap to nudge towards partial cooperation and expensive to push towards certainty (Figure 5). At Nemotron's $\lambda = 0.154$:

- 50% cooperation costs $\tau = 1.5$;
- the classical choice of $\tau = 2.5$, just above dominance, yields only about 54%;
- 90% requires $\tau \approx 16.2$.

8 A confirmatory test of the derived mechanism

Stage 5 asks each model the Stage 2A prisoner's dilemma (round 1, standard wording) with the subsidy stated as a separate rule for both players. There are 100 questions per level, balanced across the paraphrases, at four levels:

- $\tau = 0$: the baseline;
- $\tau = 1.5$: Corollary 1;
- $\tau = 2.5$: the classical guess;
- $\tau^* = 1.9 + \ln 9/\hat{\lambda}_m$: Theorem 3's subsidy for 90%, at model m's fitted precision; the derived mechanism.

Because $\tau^* > 2.5$ whenever $\lambda < 3.66$, "the derived subsidy yields more cooperation" would test nothing. The registered comparison is between the theories' *predictions*:

- at $\tau = 2.5$, is QRE's prediction closer to the observed rate than classical theory's 100%?
- at τ^*, is the observed rate within $90 \pm 15\%$?

Decision-gate criteria (held-out calibration) and an interpretation key for every pattern were fixed before the remaining fits existed. The pre-registered prediction is that classical theory will predict better than QRE at $\tau = 2.5$ for Nemotron and Kimi, because their low prisoner's dilemma λ reflects a cooperation norm, not insensitivity to payoffs.

[TBD: Results: cooperation by level and model; QRE vs classical at $\tau = 2.5$; the derived mechanism at τ^*; the halfway-point test.]

9 Discussion

"Derive or recall" has no single answer. The same question receives different answers across models and games:

- Nemotron's Cournot play survives the removal of every cue that could trigger recall;
- Gemini's Cournot play does not survive the removal of the game's name;
- for games whose rules are their fingerprint (the second-price auction, public goods), no wording we tried could hide the game from strong models, so derivation and recall remain observationally equivalent there.

This last point is a limit of the method, not a failure of it, and it is worth stating plainly: recognition of a game's structure is itself a form of reasoning.

Recognition can pull away from equilibrium. The usual worry is that recall makes models look more rational than they are. In the prisoner's dilemma we find the reverse. Recognising the game lowers equilibrium play: Gemini by 62 points, and Nemotron by 34 points relative to the extreme wording. The retrieved content is the folk lesson of repeated dilemmas (trust, reciprocity, establishing cooperation), not the backward-induction solution. For mechanism design this matters directly: an agent's behaviour depends on which famous game the designer's rules resemble.

Probes change conclusions. Without competence probes, Gemini's disguised auction would read as a 92-point strategic collapse. With them, it is correctly a reading failure. Without a registered slip rule, Nemotron's extreme-wording Cournot and auction results would read as 46-

and 51-point drops. Both safeguards altered headline numbers, which is the strongest argument for building them in before the data.

Implications for designing mechanisms for AI agents. Theorem 1 gives a cheap diagnostic: rescale the payment rule and check whether bids rescale. Theorem 3 shows that noise makes high compliance expensive, so subsidies should be set from measured responsiveness rather than from dominance thresholds. And the Stage 2A results warn that a fitted λ can encode a norm rather than noise: a low prisoner's-dilemma precision may reflect cooperative dispositions that respond sharply to incentives. Stage 5 tests exactly this.

10 Limitations

- **Disguises can fail.** The standard disguise did not hide the games from two of three models, and no wording hid the auction or public goods. Recognition is coded from written reasoning, so silent recognition cannot be excluded.
- **A leaky answer box.** In Stage 2A's disguised wording the answer field still carried the named game's field names ("bid", "quantity", "contribution"). This was found after collection, is recorded as deviation 8, and was removed in Stage 2C.
- **Timing.** Stage 2C was collected weeks after Stage 2A, so a provider updating a model in between would appear as a difference.
- **The behaviour model.** QRE treats every departure from equilibrium as payoff-scaled noise. Our models' departures are often slips or norms instead, so λ is a summary, not a structural parameter [14], and it is never compared across games.
- **The stage-game QRE in the prisoner's dilemma.** It ignores continuation play, which is harmless when the stage equilibrium is unique but not at $\tau = 1.5$.
- **Scope.** Four models, four games and free API tiers with default sampling. Cross-model patterns are rankings, not estimates of population effects.

11 Conclusion

Whether an LLM agent derives or recalls a game's equilibrium is an empirical property of the model and the game, and it can be measured. Isomorphic twins with same-wording competence probes, measured recognition and an extreme disguise separate the two where the game can be hidden at all. On that evidence, one model solves a Cournot game it no longer recognises, while another loses the Cournot equilibrium when the name goes. In the prisoner's dilemma, recognising the game pulls models away from equilibrium, towards cooperation. For mechanism design, these are not curiosities: they determine whether a rule tuned to observed play will survive a change in the rules. [TBD: one sentence on Stage 5 once available.]

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Data and code availability

Code, prompts, seeds, raw answers, analysis outputs and the locked analysis plans are available at [TBD: public repository URL]. Pre-registrations: osf.io/2uqpj, osf.io/6hz8d, osf.io/cqnsy.

References

[1] Alan Agresti and Yongyi Min. Simple improved confidence intervals for comparing matched proportions. *Statistics in Medicine*, 24(5):729–740, 2005.

[2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. Playing repeated games with large language models. *Nature Human Behaviour*, 2025. doi: 10.1038/s41562-025-02172-y.

[3] Jean-Pierre Benoît and Vijay Krishna. Finitely repeated games. *Econometrica*, 53(4):905–922, 1985.

[4] Jinhao Duan et al. GTBench: Uncovering the strategic reasoning capabilities of LLMs via game-theoretic evaluations. In *Advances in Neural Information Processing Systems*, 2024.

[5] Caoyun Fan, Jindou Chen, Yaohui Jin, and Hao He. Can large language models serve as rational players in game theory? a systematic analysis. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 17960–17967, 2024. doi: 10.1609/aaai.v38i16.29751.

[6] Jacob K. Goeree, Charles A. Holt, and Thomas R. Palfrey. *Quantal Response Equilibrium: A Stochastic Theory of Games*. Princeton University Press, 2016.

[7] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[8] John J. Horton, Apostolos Filippas, and Benjamin S. Manning. Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research, 2023.

[9] Jingru Jia, Zehua Yuan, Junhao Pan, Paul E. McNamara, and Deming Chen. LLM strategic reasoning: Agentic study through behavioral game theory. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2502.20432.

[10] R. Thomas McCoy, Shunyu Yao, Dan Friedman, Matt Hardy, and Thomas L. Griffiths. Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences*, 121(41):e2322420121, 2024. doi: 10.1073/pnas.2322420121.

[11] Richard D. McKelvey and Thomas R. Palfrey. Quantal response equilibria for normal form games. *Games and Economic Behavior*, 10(1):6–38, 1995.

[12] Roger B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.

[13] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.

[14] Mateo Pechon-Elkins and Jon Chun. Auditing game-theoretic measures of strategic reasoning in LLMs, 2026. arXiv:2603.10029.

[15] Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory.* Cambridge University Press, 2016.

[16] William Vickrey. Counterspeculation, auctions, and competitive sealed tenders. *The Journal of Finance*, 16(1):8–37, 1961.

A Deviations from the pre-registrations

1. **PD scoring error in practice runs.** Named answers were scored against the disguised labels; fixed before Stage 2.
2. **Gemini 3.8 Flash replaced** by Gemini 3.5 Flash Lite, because of a 20-request daily cap.
3. **Empty replies** (bursts of "!!!!") are re-asked rather than scored as unreadable.
4. **gpt-oss's daily token cap** slows its collection; this does not affect the design.
5. **A 10-minute NVIDIA access outage.** Its 4,454 refused requests were relabelled as infrastructure failures and re-asked.
6. **The fraction slip.** Recorded as an exploratory observation, and corrected only in the pre-registration-3 comparison.
7. **Block D (reasoning effort)** was dropped before any of its data existed, so that all models answer the same questions.
8. **The disguised answer boxes** carried the named game's field names (a limitation).
9. **Ten questions re-asked** that should have been scored as the model's own (1 Kimi Stage 2A; 2 Nemotron Stage 2A; 7 Nemotron Stage 2B traps). The re-asked answers stand.
10. **The pilot answer form** sent explanatory text that leaked words; fixed before Stage 2C.

B Stage 2A precision fits

Nemotron's Stage 2A Cournot fit is dominated by uncorrected fraction slips (answers such as 683 for 683/60), which are scored as they were written, as pre-registration 1 requires. With the slip correction fixed in pre-registration 3, its Cournot λ is 0.0082 $[0.0063, 0.0103]$ (Table 3). Kimi's Cournot difference arises entirely within the ± 0.5 tolerance: both wordings are at 100% equilibrium play.

Table 4: Posterior λ (median and 95% interval), named vs disguised. "$\geq$" marks a cell at the ceiling of the grid, where only a lower bound is identified. λ is on each game's payoff scale and is not comparable across games.

Model	Game	λ named	λ disguised	$\Delta\lambda$
Nemotron	Vickrey	≥ 944	≥ 944	both at ceiling
	Cournot	0.0012	0.0010	$+0.0001$, $P(>0) = 0.82$
	PD	0.154 $[0.002, 0.47]$	0.188 $[0.002, 0.50]$	-0.02, $P(>0) = 0.44$
	Public goods	≥ 21.8	≥ 21.8	both at ceiling
Kimi	Vickrey	≥ 944	≥ 944	both at ceiling
	Cournot	≥ 299	19.4 $[14.1, 26.6]$	$\geq +272$
	PD	0.0075 $[0.001, 0.11]$	0.079 $[0.001, 0.39]$	-0.06, $P(>0) = 0.21$
	Public goods	2.51 $[2.11, 2.90]$	3.98 $[3.35, 4.73]$	-1.51, $P(>0) = 0.00$
Gemini	Vickrey	≥ 944	3.98 $[1.5, 7.7]$ $(n = 11)$	$\geq +936$
	Cournot	0.649 $[0.50, 0.82]$	0.0047 $[0.004, 0.006]$	$+0.645$, $P(>0) = 1.00$
	PD	0.0032 $[0.001, 0.023]$	0.631 $[0.33, 0.97]$	-0.629, $P(>0) = 0.00$
	Public goods	≥ 21.8	0.434 $[0.36, 0.53]$	$\geq +21.2$